

QQQ Tactical Action

Strategy Specification · Locked Configuration · **Revision 3.0** · July 2026 · supersedes Rev 2.2 · post-audit

19.2%	0.77	-42%	-60%	~16
CAGR 1986–2026	SHARPE	REALIZED MAX DD	PLAN-FOR DRAWDOWN	TRADES / YEAR

A two-layer tactical overlay on the Nasdaq-100. A macro **fragility monitor** decides *whether* to hold equity risk; a **daily volatility-scaled leverage overlay** decides *how much*. The monitor converts buy-and-hold QQQ's -83% drawdown into -42% while beating its return; the overlay adds a modest return increment at unchanged risk. Revision 2.0 incorporates an extensive robustness review: corrected benchmark conventions, a three-window out-of-sample decomposition, a full reproducibility recipe, point-in-time data analysis, and four tested-and-rejected enhancements.

1 · Design philosophy

MACRO GATE — PERMISSION Slow valuation and credit signals (CAPE, volatility suppression, credit-spread compression) plus a trigger gauge. The only layer that can move to cash. Answers: is equity risk safe to hold?	DAILY OVERLAY — THROTTLE Price trend and realized volatility. Floors at 100% QQQ and only adds leverage above 1×. Answers: given permission, how much exposure? Never de-risks below fully invested — hence "volatility-scaled leverage overlay," not volatility targeting.
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2 · State machine

► Monitor risk-off → 100% SGOV. Master override.
► Risk-on + daily $T \geq 0.60$ → 100% QQQ, no leverage.
► Risk-on + daily $T < 0.60$ + $\text{price} > 200\text{-day}$ + $\text{slope} > 0.03$ → vol-scaled leverage , floored 1×, capped 2× via no-margin QQQ/QLD blend.

Risk-off: prior completed month's $V \geq 0.54$ and $T \geq 0.70$, and price closes below its 200-day. Re-enter after **15 consecutive** closes above. **Leverage:** exposure = clip(0.20 / realized_vol, 1.0, 2.0); for exposure E hold (2-E) QQQ and (E-1) QLD. Daily-T gate debounced: cut at 0.60, restore below 0.50.

3 · Locked parameters

PARAMETER	VALUE	PARAMETER	VALUE
Target volatility	0.20	Danger — vulnerability	$V \geq 0.54$
Leverage cap / floor	2.0× / 1.0×	Danger — trigger	$T \geq 0.70$
Volatility floor / window	0.10 / 20d	Leverage cut / restore	0.60 / 0.50
Slope window / deadband	21d / 0.03	Trend gate	200-day MA
Rebalance band	10%	Re-entry	15 closes above
Execution	next close	Cash vehicle	SGOV / T-bill

Volatility-target scaling note. tv = 0.20 is calibrated to QQQ's own median realized volatility (18.3%), giving ≈1.09× exposure at typical conditions. The transferable parameter is **tv ÷ median volatility**, not tv itself. Applying this architecture to another index requires re-solving: for the S&P 500 (median vol 12.7%) the equivalent is tv ≈ 0.14. Using 0.20 on the S&P produces 1.31–1.42× average exposure — a materially more aggressive strategy.

4 · Results — apples-to-apples

All series computed identically — **total return**, a **single** next-close execution lag, 5 bps per one-way turnover, 1986–2026:

STRATEGY	CAGR	SHARPE	SORTINO	MAX DD	\$1 →
Integrated (tv 0.20)	19.2%	0.77	0.93	-42%	\$1,250
Fragility monitor alone	17.9%	0.78	0.94	-40%	\$803
Buy & hold QQQ	14.9%	0.56	0.75	-83%	\$282

Rev 3.0 corrections (independent audit, July 2026). An external audit of the Rev 2.2 reference implementation found two convention errors, both of which *understated* the strategy by a combined ~1.0% CAGR. **(1) Double execution lag.** The published pos is the position *held* on day t — it equals the raw state-machine signal shifted once. Rev 2.x lagged it again. Diagnostic: on position-change days, same-day pos reproduces the published *st ret* with MAE 9.4e-06; lagging again gives 1.17e-02. **(2) Return basis.** The strategy used the price series px while the benchmark bhqqq is total return (~0.60%/yr of dividends). Both are now total return. Earlier corrections retained: Rev 1.0 quoted a price-return benchmark (14.2%); Rev 2.2 found that terminal wealth, drawdown, and COVID figures had been computed without the 10% rebalance band that the spec listed as locked — which also reconciled an external observation of ~65 exposure changes per year (the unbanded target series) against the stated ~16 (the banded executed series).

5 · Regime behavior

REGIME	INTEGRATED	BUY-HOLD	BEHAVIOR
Dot-com 2000–02	–36%	–83%	Monitor in cash; leverage dormant
GFC 2007–09	–14%	–53%	Exited Feb 2008 on January's frozen gauge
2022 bear	–18%	–35%	Partial exit
COVID 2020	–31%	–28%	Too fast; never exited
Black Monday 1987	–42%	–40%	One-day crash; unlevered

WHY THE MONITOR CAUGHT THE GFC

The exit fired in February 2008 using **January's** frozen reading (V 0.553, T 0.945) — even though February's own V had already fallen to 0.537, below threshold. Because CAPE is price-over-earnings, **V collapses during a crash** (0.553 → 0.236 → 0.158 across the GFC). The prior-month freeze preserves the pre-crash valuation long enough to fire. A faster-updating V would have disarmed the gate exactly when it was needed.

6 · Out-of-sample validation

6.1 · S&P 500 three-window decomposition

The identical architecture applied to the S&P 500. Only 1929–1985 is genuinely non-overlapping in time; the 1986–2026 portion shares crash observations with the QQQ development sample.

WINDOW	BUY & HOLD	200-DAY TREND ONLY	MONITOR	INTEGRATED
1929–1985 (true OOS)	3.9% / –86%	7.8% / –48%	7.3% / –57%	7.5% / –57%
1986–2026 (in-sample)	9.2% / –57%	8.1% / –22%	11.3% / –34%	12.0% / –36%
1929–2026 (combined)	6.1% / –86%	7.9% / –48%	8.9% / –57%	9.4% / –57%

HONEST READING

The architecture's crash defense replicates convincingly out-of-sample — the Great Depression was cut from –86% to –36%. However, in the true OOS window the **V/T monitor does not outperform a plain 200-day trend rule** (Sharpe 0.27 vs 0.32 on a like-for-like state machine, or 0.27 vs 0.29 with identical 15-day re-entry). The monitor's advantage over trend-following appears decisively in modern data on both indexes (QQQ 0.63 vs 0.51; S&P 0.57 vs 0.37) but is unconfirmed pre-1985. This may reflect genuine regime dependence or thin pre-1950 credit data; it cannot be resolved from available history.

6.2 · Monte Carlo — 2,000 resampled 40-year histories

Method: 63-day block bootstrap of the **input panel** (returns, monitor state, daily T, cash rate), with the **full state machine re-run on each synthetic path** — not a resampling of finished strategy returns.

PERCENTILE	1ST	5TH	25TH	50TH	75TH	90TH	99TH
CAGR	9.6%	11.9%	15.4%	18.0%	20.6%	22.9%	26.9%
Max drawdown	–68%	–60%	–51%	–45%	–41%	–36%	–30%

Realized 18.4% sits at the bootstrap **median** (18.0%) — typical, not fortunate. But the realized –42% drawdown sits on the *favorable* side: median is –45%, 5th percentile –60%, 1st percentile –68%.

6.3 · Worst-possible-timing & walk-forward

Across every historical start date the **worst 10-year outcome was +5.7%/year** (from the March 2000 peak, where buy-and-hold lost –7.7%/year). No start date produced a negative 10-year result. Holdout test (2016–2026, rules not provably frozen beforehand): 22.1% / 0.91 Sharpe.

7 · What V actually does

Extensive testing established that the vulnerability gauge is **not a crash predictor**. V exceeded 0.54 at 76% of major drawdown peaks versus a 64% unconditional base rate — a lift of 1.19× on 17 observations. Its threshold is not binding: 13 of 14 QQQ drawdown peaks cleared 0.54 comfortably (median 0.79). T rises from ~0.39 at the peak to ~0.78 at the exit and does the timing.

V'S ACTUAL FUNCTION: A BOTTOM FILTER

V distinguishes "expensive market breaking down" from "cheap market thrashing." T and the 200-day cannot tell these apart — they look identical on price and volatility. Testing a relaxed V threshold during post-crash periods (to catch secondary declines like 1938) produced **five additional exits, all headfakes** (1934, 1939, 1940, 1987, 2010; markets rose 7.9%–18.3% while out), a combined multiple of 0.569×, and a *deeper* drawdown (–52% → –60%). V's collapse after a crash is protective, not a defect.

7.1 · What the gate is actually for: filtering 200-day breaks

The clearest measure of the monitor's contribution is not crash prediction but **which trend breaks are worth trading**. Every episode where price closed below its 200-day average, classified by whether the V/T gate was armed:

SAMPLE	GATE STATE	EPISODES	MARKET LOWER AT RE-ENTRY	AVG DROP TO TROUGH	MULTIPLE IF EXITED
QQQ 1986–2026	ARMED (strategy exits)	10	3 / 10	–21.5%	3.215×
	not armed (stays invested)	35	1 / 35 (3%)	–4.8%	0.053×
S&P 1929–2026	ARMED	29	13 / 29	–17.1%	12.887×
	not armed	74	7 / 74 (9%)	–4.7%	0.040×

Across 103 S&P trend breaks in 97 years the gate acts on 29 and ignores 74. Acting on the 29 multiplies capital 12.9×; acting on the 74 would multiply it by 0.04×. Exiting on *every* break — pure trend-following — produces 0.17× on QQQ before cash interest. **The 200-day break is an extremely noisy signal; the V/T gate is what makes it tradeable.** This also reconciles §6.1: in the 1929–1985 window the trend rule and the monitor reach similar Sharpe ratios by opposite routes — the trend rule trades all 103 breaks and collects T-bills through the bad ones, while the monitor trades 29 and stays invested through the rest. **Known exposure:** four of seventeen S&P declines >20% since 1929 began with V below threshold (1933, 1938, 1948, 1980), the worst being –46% in 1938 requiring 4.1 years to recover. None since 1980. The characteristic pattern is a secondary decline after valuations have already been compressed. The model should be expected to provide no protection in that scenario.

8 · Point-in-time data integrity

Five of six scored gauges derive from daily index prices and daily FRED corporate yields — same-day availability, never revised. **Only CAPE carries publication-lag and revision exposure.**

TEST	CAGR	MAX DD
Baseline	18.7%	–42%
CAPE lagged 1 / 3 / 6 / 12 months	17.9–18.1%	–42%

CAPE enters as a percentile rank of a 120-month earnings average, which damps revisions structurally: a 20% revision to one quarter moves CAPE ≈0.5%, shifting its rank <0.005 on average and altering the $V \geq 0.54$ condition in under 1% of months. Drawdown is unchanged under every lag. **The lag effect is concentrated in three near-threshold decisions** (Aug 1990, Aug 2011, Oct 2022 — gaps to threshold of 0.007, 0.021, 0.005), which together account for ~0.8% of CAGR. V sits within ±0.02 of threshold in 8.2% of months; forward results should be expected to vary comparably based on how future marginal calls resolve.

Unresolvable residual: Shiller's historical CAPE reflects decades of accumulated revisions and has no public vintage archive, so the pre-1985 out-of-sample test uses a version of history unavailable at the time. Sensitivity testing is the only available evidence that this does not matter materially.

9 · Reproducibility recipe

The production position series is exactly reproducible from published files, verified at **100.00% agreement across all 10,214 trading days**:

- ▶ Take monthly `V`, `T` from `data.json` → `timeline_qqq`; resample to month-start and **shift one month** (acting gate).
- ▶ `Danger = V ≥ 0.54 AND T ≥ 0.70`. Exit when danger and `px < ma`. Re-enter after 15 consecutive closes with `px ≥ ma`.
- ▶ **Shift the resulting position series forward one day** — this is the next-close execution convention embedded in the published `pos`.

10 · Tested and rejected

Excess CAPE Yield (rate-adjusted valuation) — retains 3.5× more variance and is far less saturated than CAPE, but gate multiple falls 2.674 → 1.601. Catches the 1990 decline, which was a whipsaw that lost 24%.

Yield-curve slope (GS10–TB3MS, 1953–2026) — 20% of inverted months preceded a >20% drawdown within 12 months vs a 16% base rate (1.25× lift, n=13). No relationship between inversion duration and outcome; lead times ranged 8 to 89 months. The longest inversion on record (Nov 2022–Nov 2024, 25 months) produced no drawdown.

Household equity allocation (Z.1, 1945–2026) — best standalone predictor tested (1.98× lift at the 95th percentile vs CAPE's 1.65×), but **0.924 correlated with CAPE**. Substituting changes nothing; adding it as a fourth gauge degrades the gate multiple 2.13 → 1.34. *Current reading 45.8% is the highest in the entire record, exceeding the 2000 peak (38.7%) and 2022 (41.1%).*

Sub-1× volatility targeting (floor 0.25) — genuine vol-timing skill (at identical average exposure, drawdown –59% → –37%), but against the production monitor it is Sharpe-neutral (0.74 vs 0.75) and costs 1.8% CAGR. Bootstrap: better drawdown in 96% of paths, worse CAGR in 97%. Same efficient frontier, lower risk point. Rejected in favor of the higher-return point given a <10% portfolio allocation.

Headfake reduction — three approaches, all rejected. Six of nine QQQ exits (15 of 24 on the S&P) were headfakes. (a) *Ex-ante identification*: no feature separates good exits from headfakes. Drawdown-from-high scores 0.89 separation on QQQ but 0.42 on the S&P; T scores 0.73 on the S&P but 0.43 on QQQ; every candidate feature shows complete range overlap in both samples. (b) *Partial de-risking* (holding 25–75% equity instead of full SGOV): worse on every metric — at 50% equity the dot-com drawdown widens from –36% to –62%. (c) *Faster re-entry* (1–10 days instead of 15): materially worse — one-day re-entry produces a –68% drawdown by re-entering into the decline. **The headfake cost and the crash protection are the same mechanism.** Because good exits average –40% while headfakes average +13%, any proportional reduction gives back more than it saves.

Post-crash V relief — relaxing V after a >30% drawdown. All five added exits were headfakes; combined multiple 0.569×; drawdown worsened to –60%.

11 · Known limitations

READ BEFORE COMMITTING CAPITAL

- **Plan for –60%, not –42%.** Bootstrap 5th percentile is –60%, 1st percentile –68%, worst of 2,000 paths –84%.
- **Fast crashes are the structural blind spot.** 1987 and COVID outran every signal. T was 0.36 and 0.21 at those peaks and never rose in time.
- **The gate is wrong most of the time, and this cannot be improved.** Six of nine QQQ exits were headfakes; three separate approaches to reducing them (prediction, partial exits, faster re-entry) each made results worse. Expect to sit in SGOV watching the market rally in roughly two of every three exits. That is the premium paid for the one-in-three that matters, and every tested method of lowering the premium cost more than it saved.
- **V/T complexity is unconfirmed out-of-sample.** See §6.1. Decisive in modern data, a tie pre-1985.
- **CAPE is a decaying input.** Its rank variance fell from 0.298 (1929–90) to 0.060 (2010–26). Replacing it with a constant costs 2.1% CAGR and no drawdown — the plausible path of further degradation.
- **Hypothetical history.** Pre-1999 QQQ is a price index; pre-2006 QLD is synthetic; pre-2020 SGOV is proxied. Not achievable returns.
- **Not live-validated.** All results are historical. Paper-trading is required before capital.

13 · Reference implementation

A self-contained Python script (`reference_backtest.py`, pandas/numpy only, no network calls) regenerates every figure above from the two published JSON files. It prints an integrity check confirming that the raw state-machine signal, shifted once, reproduces the published pos at **100.00% across all 10,216 trading days**.

Conventions, stated explicitly in code: one execution lag applied once to the composite target; the published pos treated as already-executed; total return throughout for strategy and benchmark alike; the no-margin QQQ/QLD/SGOV weighting with a runtime assertion that weights sum to 1.0; synthetic QLD across the whole sample; Sharpe on excess return over the recovered cash series; the 10% rebalance band applied to the target before execution; and configurable one-way transaction costs. The one flagged approximation is the cash rate, recovered from the ~15% of days the monitor was risk-off and forward-filled — substituting a flat 2% changes CAGR by less than 0.1%.

Audit trail. Revision 3.0 exists because an independent reviewer ran a test this document had not: comparing `stret` against same-day versus lagged pos on position-change days. Both conventions produce a 100% match on the *position series*; only the return reconstruction distinguishes them. The reproducible artifact is what made the error findable.

12 · Operations

Check the dashboard after each close; it displays the target for the next trading day. Cadence ≈16 trades/year and ~9 risk-off round-trips per 40 years.

Execution timing is not a material risk: delaying execution a full extra day costs 0.2% CAGR with unchanged drawdown; a four-day delay costs 0.4%. Slippage at 5 bps/trade costs 0.23% CAGR — on penny-spread instruments, effectively free. **Forward-state display.** The dashboard computes the monitor state implied by *today's* close rather than reading the published pos (which is the position held today, decided at the prior close). Without this the displayed

target would be one session stale on the 18 monitor-transition days in the record \u2014 precisely the risk-off and re-entry moments. **Data-integrity guards.** The dashboard blocks its recommendation and warns if either feed is more than three business days stale, if the feeds disagree by more than two days, or if the feeds are unreachable (the offline snapshot is explicitly marked non-tradeable). It also detects and discards an **incomplete final bar** — if the last row carries the current date and it is before the 4:00pm ET close, that partial bar is excluded and the last completed session is used instead. The scheduled pipeline runs at 22:00 UTC, safely after the close; the guard covers manual mid-session runs, which would otherwise publish a partial bar and corrupt the volatility, slope, and 200-day comparisons.

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